Supplemental material:

1. **Search terms**

	Database(s)	Search Term
1	Medline	(antipsychotic).ti,ab
2	Medline	(major tranquilizer).ti,ab
3	Medline	exp "ANTIPSYCHOTIC AGENTS"/
4	Medline	exp MOTIVATION/
5	Medline	exp REWARD/ OR exp "DELAY DISCOUNTING"/
6	PsycINFO	exp "NEUROLEPTIC DRUGS"/
7	PsycINFO	exp INCENTIVES/ OR exp REWARDS/ OR exp MOTIVATION/
8	PsycINFO	exp "MONETARY REWARDS"/ OR exp "MONETARY INCENTIVES"/
9	EMBASE	exp "NEUROLEPTIC AGENT"/
10	EMBASE	(incenti*).ti,ab
11	EMBASE	(rewar*).ti,ab
14	PsycINFO	(7 OR 8)
15	EMBASE	(10 OR 11)
17	PsycINFO	(6 AND 14)

18	EMBASE	(9 AND 15)
19	Medline	(1 OR 2 OR 3)
20	Medline	(4 OR 5)
21	Medline	(19 AND 20)
22	Medline	(19 AND 20) [Human age groups Adult] [Humans]
23	EMBASE	(9 AND 15) [Human age groups Adult 18 to 64 years OR Aged 65+ years]
24	EMBASE	(9 AND 15) [Human age groups Adult 18 to 64 years] [Humans]
25	PsycINFO	(6 AND 14) [Human age groups Adulthood 18 Yrs + Older] [Population Human]

2. Prisma checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Title
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	n/a short abstract
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	P5-7
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	P7
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	P7
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	P7

Section and Topic	Item #	Checklist item	Location where item is reported
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Suppl 3
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	P8
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	P8
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	P8
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	P8
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	P8
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	n/a
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	P8
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	n/a
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	P8
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	P8
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	P9
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	N/a
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	n/a
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	P18
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	P11
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	P11
Study characteristics	17	Cite each included study and present its characteristics.	Tables 1, 3, 4, and 5

Section and Topic	Item #	Checklist item	Location where item is reported
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Table4 and suppl 6
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Table 2 and 4
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	P18
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Table 1, 4, and 5
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	n/a
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	n/a
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	P18
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Table 5
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	P15
	23b	Discuss any limitations of the evidence included in the review.	P18
	23c	Discuss any limitations of the review processes used.	P18
	23d	Discuss implications of the results for practice, policy, and future research.	P18-19
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	P7
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	P7
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	P7
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	P9
Competing interests	26	Declare any competing interests of review authors.	Title page
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	N/a

3. ENTREQ Table

No	Item	Guide and description	Page
1	Aim	State the research question the synthesis addresses.	P7
2	Synthesis methodology	Identify the synthesis methodology or theoretical framework which underpins the synthesis, and describe the rationale for choice of methodology *(e.g. meta-ethnography, thematic synthesis, critical interpretive synthesis, grounded theory synthesis, realist synthesis, meta-aggregation, meta-study, framework synthesis).*	P8
3	Approach to searching	Indicate whether the search was pre-planned (*comprehensive search strategies to seek all available studies)* or iterative (*to seek all available concepts until they theoretical saturation is achieved)*.	P7-8
4	Inclusion criteria	Specify the inclusion/exclusion criteria *(e.g. in terms of population, language, year limits, type of publication, study type).*	P7
5	Data sources	Describe the information sources used (e.g. *electronic databases (MEDLINE, EMBASE, CINAHL, psycINFO, Econlit), grey literature databases (digital thesis, policy reports), relevant organisational websites, experts, information specialists, generic web searches (Google Scholar) hand searching, reference lists)* and when the searches conducted; provide the rationale for using the data sources.	P7
6	Electronic Search strategy	Describe the literature search *(e.g. provide electronic search strategies with population terms, clinical or health topic terms, experiential or social phenomena related terms, filters for qualitative research, and search limits).*	P7
7	Study screening methods	Describe the process of study screening and sifting *(e.g. title, abstract and full text review, number of independent reviewers who screened studies).*	Fig 1, p8
8	Study characteristics	Present the characteristics of the included studies *(e.g. year of publication, country, population, number of participants, data collection, methodology, analysis, research questions).*	Table 1 and 2 and suppl 6
9	Study selection results	Identify the number of studies screened and provide reasons for study exclusion *(e,g, for comprehensive searching, provide numbers of studies screened and reasons for exclusion indicated in a figure/flowchart; for iterative searching describe reasons for study exclusion and inclusion based on modifications t the research question and/or contribution to theory development).*	Fig 1

10	Rationale for appraisal	Describe the rationale and approach used to appraise the included studies or selected findings *(e.g. assessment of conduct (validity and robustness), assessment of reporting (transparency), assessment of content and utility of the findings).*	P18
11	Appraisal items	State the tools, frameworks and criteria used to appraise the studies or selected findings *(e.g. Existing tools: CASP, QARI, COREQ, Mays and Pope* [25]*; reviewer developed tools; describe the domains assessed: research team, study design, data analysis and interpretations, reporting).*	P8
12	Appraisal process	Indicate whether the appraisal was conducted independently by more than one reviewer and if consensus was required.	P8
13	Appraisal results	Present results of the quality assessment and indicate which articles, if any, were weighted/excluded based on the assessment and give the rationale.	P18 and table 4 and 7
14	Data extraction	Indicate which sections of the primary studies were analysed and how were the data extracted from the primary studies? *(e.g. all text under the headings "results /conclusions" were extracted electronically and entered into a computer software).*	P8 and 9
15	Software	State the computer software used, if any.	P8 and 9
16	Number of reviewers	Identify who was involved in coding and analysis.	P8 and 9
17	Coding	Describe the process for coding of data *(e.g. line by line coding to search for concepts).*	P8
18	Study comparison	Describe how were comparisons made within and across studies *(e.g. subsequent studies were coded into pre-existing concepts, and new concepts were created when deemed necessary).*	P8 and 9
19	Derivation of themes	Explain whether the process of deriving the themes or constructs was inductive or deductive.	P8 and 9
20	Quotations	Provide quotations from the primary studies to illustrate themes/constructs, and identify whether the quotations were participant quotations of the author's interpretation.	Table 3
21	Synthesis output	Present rich, compelling and useful results that go beyond a summary of the primary studies (e.g. *new interpretation, models of evidence, conceptual models, analytical framework, development of a new theory or construct).*	Table 4

4. PICOS Table

Population	People with serious mental illness requiring antipsychotic treatment
Intervention	Incentives
Comparison	No Incentives
Outcome	Adherence to antipsychotic treatment
Study type	No restrictions

5. List of included papers

1. Classen D, Fakhoury WK, Ford Rm Priebe S. Money for medication: financial incentives to improve medication adherence in assertive outreach. Psychiatric Bulletin (2007), 31, 4-7
2. Staring AB, Mulder CL, Priebe S. Financial incentives to improve adherence to medication in five patients with schizophrenia in the Netherlands. Psychopharmacol Bull. 2010;43(1):5-10.
3. Priebe S, Yeeles K, Bremner S, Lauber C, Eldridge S, Ashby D, David AS, O'Connell N, Forrest A, Burns T. Effectiveness of financial incentives to improve adherence to maintenance treatment with antipsychotics: cluster randomised controlled trial. BMJ. 2013 Oct 7;347:f5847. doi: 10.1136/bmj.f5847.
4. Noordraven EL, Wierdsma AI, Blanken P, Bloemendaal AF, Staring AB, Mulder CL. Financial incentives for improving adherence to maintenance treatment in patients with psychotic disorders (Money for Medication): a multicentre, open-label, randomised controlled trial. Lancet Psychiatry. 2017 Mar;4(3):199-207. doi: 10.1016/S2215-0366(17)30045-7.
5. Financial Incentives to Improve Acceptance of Antipsychotic Injections. NCT03192631 Cochrane Central Register of Controlled Trials. 31 May 2018. Issue 5
6. Priebe S, Bremner SA, Pavlickova H. Discontinuing financial incentives for adherence to antipsychotic depot medication: long-term outcomes of a cluster randomised controlled trial. BMJ Open. 2016;6(9):e011673. Published 2016 Sep 21. doi:10.1136/bmjopen-2016-011673
7. Pavlickova H, Bremner SA, Priebe S. The effect of financial incentives on adherence to antipsychotic depot medication: does it change over time? J Clin Psychiatry. 2015 Aug;76(8):e1029-34. doi: 10.4088/JCP.14m09669. PMID: 26335089.
8. Szmukler G. Financial incentives for patients in the treatment of psychosis. J Med Ethics. 2009 Apr;35(4):224-8. doi: 10.1136/jme.2008.027151.

9. Szmukler G. Rapid response to: Effectiveness of financial incentives to improve adherence to maintenance treatment with antipsychotics: cluster randomised controlled trial. BMJ 2013;347:f5847
10. Kendall T. Paying patients with psychosis to improve adherence. BMJ. 2013 Oct 22;347:f5782. doi: 10.1136/bmj.f5782. PMID: 24149817.
11. Marteau TM, Ashcroft RE, Oliver A. Using financial incentives to achieve healthy behaviour. BMJ. 2009 Apr 9;338:b1415. doi: 10.1136/bmj.b1415. PMID: 19359291.
12. Guinart D, Kane JM. Use of Behavioral Economics to Improve Medication Adherence in Severe Mental Illness. Psychiatr Serv. 2019 Oct 1;70(10):955-957. doi: 10.1176/appi.ps.201900116.
13. Guinart D, Kane JM. Incentivizing Is Not Coercing: A Commentary. Psychiatr Serv. 2020 Jan;71(3)DOI:10.1176/appi.ps.201900543
14. Peterson-Dana C. Shared Decisions and Agency: Better Engagement Tools. Psychiatr Serv. 2019 Oct 1;70(10):857. doi: 10.1176/appi.ps.701002. PMID: 31570080.
15. Burns T. Is it acceptable for people to be paid to adhere to medication? Yes. BMJ. 2007 Aug 4;335(7613):232. doi: 10.1136/bmj.39286.399514.BE. PMID: 17673764; PMCID: PMC1939784.
16. Shaw J. Is it acceptable for people to be paid to adhere to medication? No. BMJ. 2007 Aug 4;335(7613):233. doi: 10.1136/bmj.39286.422639.BE.
17. Priebe S, Burton A, Ashby D, Ashcroft R, Burns T, David A, Eldridge S, Firn M, Knapp M, McCabe R. Financial incentives to improve adherence to anti-psychotic maintenance medication in non-adherent patients - a cluster randomised controlled trial (FIAT). BMC Psychiatry. 2009 Sep 28;9:61. doi: 10.1186/1471-244X-9-61.
18. Noordraven EL, Audier CH, Staring AB, Wierdsma AI, Blanken P, van der Hoorn BE, Roijen LH, Mulder CL. Money for medication: a randomized controlled study on the effectiveness of financial incentives to improve medication adherence in patients with psychotic disorders. BMC Psychiatry. 2014 Dec 2;14:343. doi: 10.1186/s12888-014-0343-3
19. Classen D. Financial incentives for antipsychotic depot medication: ethical issues. J Med Ethics 2007;33:189–193. doi: 10.1136/jme.2006.016188
20. Priebe S, Sinclair J, Burton A, et al. Acceptability of offering financial incentives to achieve medication adherence in patients with severe mental illness: a focus group study. *J Med Ethics*. 2010;36(8):463-468. doi:10.1136/jme.2009.035071
21. Moran K, Priebe S. Better quality of life in patients offered financial incentives for taking anti-psychotic medication: Linked to improved adherence or more money? Qual Life Res. 2016 Aug;25(8):1897-902. doi: 10.1007/s11136-016-1238-1.
22. Highton-Williamson E, Barnicot K, Kareem T, Priebe S. Offering financial incentives to increase adherence to antipsychotic medication: the clinician experience. J Clin Psychopharmacol. 2015 Apr;35(2):120-7. doi: 10.1097/JCP.0000000000000276.
23. Noordraven EL, Schermer MHN, Blanken P, Mulder CL, Wierdsma AI. Ethical acceptability of offering financial incentives for taking antipsychotic depot medication: patients' and clinicians' perspectives after a 12-month randomized controlled trial. BMC Psychiatry. 2017 Aug 29;17(1):313. doi: 10.1186/s12888-017-1485-x.
24. Noordraven EL, Wierdsma AI, Blanken P, Bloemendaal AFT, Mulder CL. The effect of financial incentives on patients' motivation for treatment: results of "Money for Medication," a randomised controlled trial. BMC Psychiatry. 2018 May 24;18(1):144. doi: 10.1186/s12888-018-1730-y.

25. Henderson C, Knapp M, Yeeles K, Bremner S, Eldridge S, David AS, et al. (2015) Cost- Effectiveness of Financial Incentives to Promote Adherence to Depot Antipsychotic Medication: Economic Evaluation of a Cluster-Randomised Controlled Trial. PLoS ONE 10(10): e0138816. doi:10.1371/journal.pone.0138816
26. Noordraven EL, Wierdsma AI, Blanken P, Bloemendaal AFT, Mulder CL. Medical and social costs after using financial incentives to improve medication adherence: results of a 1 year randomised controlled trial. BMC Res Notes. 2018 Sep 10;11(1):655. doi: 10.1186/s13104-018-3747-1.

Supplement 5: Coding Framework from Phase Two

1. **Respect for Autonomy**
   - 1.1. Risk of coercion
   - 1.2. Risk of financial dependence
   - 1.3. Disrespect for considered decisions
   - 1.4. Reduced counselling about treatment
   - 1.5. Transparency and personalisation
   - 1.6. Less restrictive option
   - 1.7. Increase in autonomy
2. **Beneficence**
   - 2.1. Increased adherence
   - 2.2. Improved outcomes
   - 2.3. Increased efficacy of other treatments
   - 2.4. May benefit compliant patients
   - 2.5. Habit formation and tolerance
   - 2.6. Limits of flexibility
   - 2.7. May not help forgetful people
   - 2.8. May not work
3. **Non-Maleficence**

3.1. Risk of perverse incentives

3.2. Increased drug and alcohol use

3.3. Risk of stigmatization of patient and antipsychotics

3.4. Increased demand for psychopharmacology

3.5. Reduced attention to adverse effects

3.6. Difficulty withdrawing incentives

4. Justice

4.1. Costs/savings for wider healthcare system

4.2. Penalizes good adherence

4.3. Exploiting power dynamic over unwell

4.4. Inclusion criteria

4.5. Inculcating a sense of entitlement

5. Others

5.1. Impact on patient dignity

5.2. Impact on intrinsic motivation

5.3. Relationship to existing reinforcement techniques

5.4. Greed, fraud and demand for more money

5.5. Difficulty setting appropriate payment levels

5.6. Loss of trust in doctor-patient relationship

5.7. Supplanting family and social support

Supplement 6: Research exploring financial incentives among people with relevant lived experience

Paper *(JBI critical appraisal score max 10)*	**Linked study**	**Method and aims**	**Participants (n)**
Classen et al 2007 (41) *(2)*	None	Questionnaires and pilot **Aims:** to investigate attitudes to and effective use of financial incentives	70 AOT leaders in England and 4 patients (7 teams had used non-financial incentives for engagement, none for adherence)
Classen 2007 (59) *(3)*	None	Analysis of free text answers for team leaders with objections **Aims:** to identify and classify major ethical concerns with money for medication.	Only the 53 of 70 team managers who had concerns
Priebe et al 2010 (60) *(8)*	No trial	25 focus groups **Aims:** to explore attitudes, concerns and opinions of relevant stakeholders	139 participants including 76 mental health staff, 27 patients, 16 carers and 20 other stakeholders eg trust NEDs health economics recruited via CMHTs, AOTs and staff
Staring et al 2010 (42) *(3)*	Staring et al pilot	A short questionnaire **Aims:** to test the feasibility of using financial incentives	All 5 patients, two mothers,

Moran & Priebe 2016 (61) *(n/a see FIAT RCT)*	FIAT	Regression models of trial data comparing adherence, money received and wellbeing **Aims:** to determine whether QOL increase was due to adherence or financial incentives	56 intervention group patients who completed DIALOG subjective QOL scores
Highton-Williamson et al 2015 (62) *(8)*	FIAT	Semistructured interviews at 6 months and 12 months and thematic analysis **Aims:** to explore clinicians' experiences of financial incentives	59 clinicians from the FIAT study
Noordraven et al 2017 (63) *(2)*	MfM	Structured interviews with 2 open questions and 19 statements to rate. **Aims:** to assess "attitudes and ethical considerations of" patients and clinicians in MfM trial	133 patients 97 clinicians (46% nurses)
Noordraven et al 2018 (64) *(n/a see MfM RCT)*	MfM	HoNOS (n=131) and TEQ (n=61) assessments of treatment motivation at 0, 12 and 18 months **Aims:** to assess impact of financial incentives on motivation	Participants from the MfM trial. At 6 months, 66 from intervention and 65 from control. At 18 months 60 and 49 respectively.

Supplement 7: Economic analyses of financial incentives for antipsychotic depot treatment

	Linked study	Included patients	Total health costs at baseline	Physical health costs during intervention	Mental health costs during intervention	Of which MH admissions during intervention	Total health costs during intervention	Social/criminal costs	Conclusion
Henderson et al (in GBP) (65)	FIAT	117/141							97.5% likelihood that "good adherence" can be achieved without these costs exceeding £27,800
Intervention		59	10088 (SD 1059)			3407 (SD 1101)	9350 (SD 1189)		
Control		78	10511 (SD 2004)			5105 (SD 1787)	8651 (SD 1890)		

Noordraven et al 2018 (in Euros) (66)	MfM	133/169	Not stated but "negligible" differences						No overall cost reduction
Intervention		69/		529.6 (SD2241.7)	1062.9 (SD3031.5)	613.3 (SD 2788.7)	1592.5 (SD 3700.7)	248.4 (SD856.2)	
Control		64/		484.0 (SD226.66)	788.8 (SD2379.3)	433.1 (SD 2379.3)	1272.8 (SD 3223.7)	229.3 (SD1477.4)	